

Fyntra is the largest online clothing company in USA. It sells clothing online, but they also have in-store style and clothing advice sessions. Customers come into the store, have sessions/meetings with a personal stylist, then can go home and order either on a mobile app or website for the clothes they want.

Company wants to decide whether to focus the effort on mobile app experience or its website. As a drastic measure it is also evaluating to shut down the website.

You as a ML expert in the team will help the company make the right decision.

Tasks to be performed: Increase in profits as the focus on the optimal sales channel will result into the higher top line and the higher bottom line.

```
#Importing necessary libraries.
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, mean_squared_error
```

```
#Importing the Dataset.
df=pd.read_csv("FyntraCustomerData.csv")
df.head()
```

	Email	Address	Avatar	Avg_Session_Length	Time_on_App	Time_on_Website	Length_of_Mer
0	mstephenson@fernandez.com	835 Frank Tunnel\nWrightmouth, MI 82180-9605	Violet	34.497268	12.655651	39.577668	
1	hduke@hotmail.com	4547 Archer Common\nDiazchester, CA 06566-8576	DarkGreen	31.926272	11.109461	37.268959	
2	pallen@yahoo.com	24645 Valerie Unions Suite 582\nCobbborough, D...	Bisque	33.000915	11.330278	37.110597	
3	riverarebecca@gmail.com	1414 David Throughway\nPort Jason, OH 22070-1220	SaddleBrown	34.305557	13.717514	36.721283	
4	mstephens@davidson-herman.com	14023 Rodriguez Passage\nPort Jacobville, PR 3...	MediumAquaMarine	33.330673	12.795189	37.536653	

Next steps:

[Generate code with df](#)
[View recommended plots](#)
[New interactive sheet](#)

```
#Checking the null values in the Dataset.
df.isnull().sum()
```

	0
Email	0
Address	0
Avatar	0
Avg_Session_Length	0
Time_on_App	0
Time_on_Website	0
Length_of_Membership	0
Yearly_Amount_Spent	0

```
#Checking the Dataset Information.
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
#   ...
```

```

---
0  Email          500 non-null  object
1  Address        500 non-null  object
2  Avatar         500 non-null  object
3  Avg_Session_Length  500 non-null  float64
4  Time_on_App    500 non-null  float64
5  Time_on_Website 500 non-null  float64
6  Length_of_Membership 500 non-null  float64
7  Yearly_Amount_Spent 500 non-null  float64
dtypes: float64(5), object(3)
memory usage: 31.4+ KB

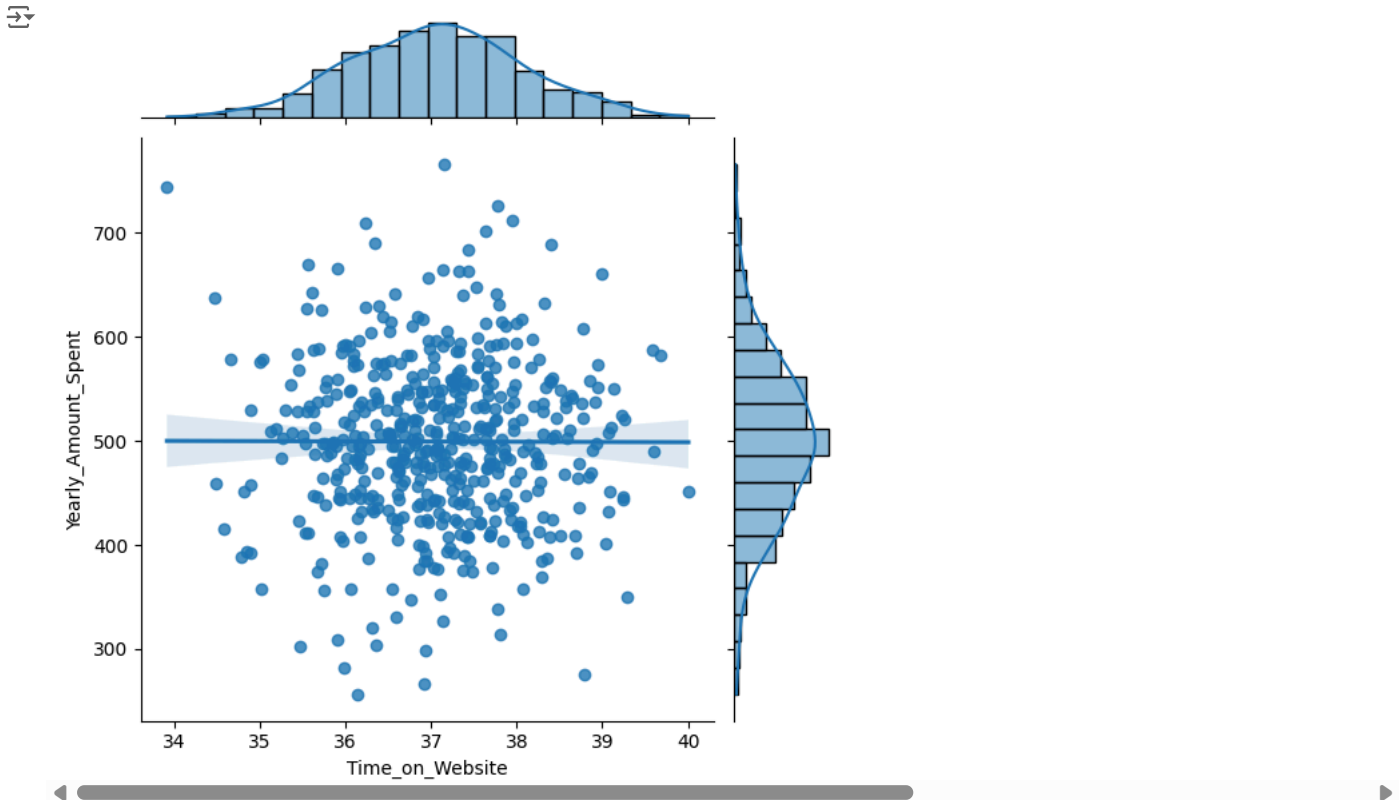
```

1. Compute -- Use seaborn to create a jointplot to compare the Time on Website and Yearly Amount Spent columns. Is there a correlation?

```

sns.jointplot(x='Time_on_Website',y='Yearly_Amount_Spent',data=df,kind='reg')
plt.show()

```



In the scatter plot, points are cluster around a straight line, there might be a correlation but it's weak

```

website_corr = df['Time_on_Website'].corr(df['Yearly_Amount_Spent'])
print(f"Correlation between Time on Website and Yearly Amount Spent: {website_corr}")

```

Correlation between Time on Website and Yearly Amount Spent: -0.002640844717963287

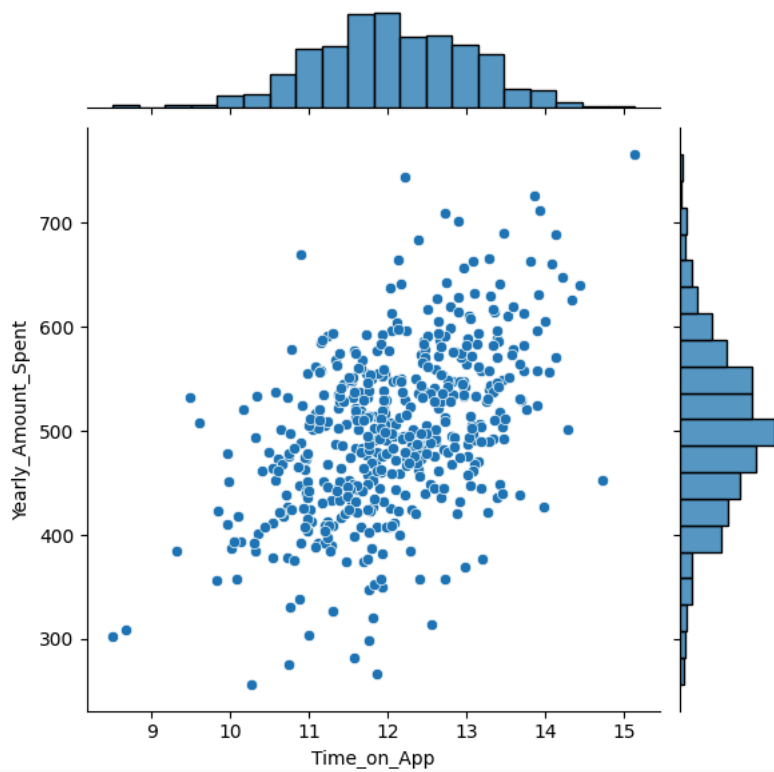
The correlation is weak (closer to 0), spending time of website might not significatnly influence yearly spending.

2. Compute – Do the same as above but now with Time on App and Yearly Amount Spent. Is this correlation stronger than 1st One?

```

sns.jointplot(x='Time_on_App',y='Yearly_Amount_Spent',data=df)
plt.show()

```



The points are moving upward. There is a correlation but it's less positive.

```
app_corr = df['Time_on_App'].corr(df['Yearly_Amount_Spent'])
print(f"Correlation between Time on App and Yearly Amount Spent: {app_corr}")
```



Correlation between Time on App and Yearly Amount Spent: 0.4993277700598346

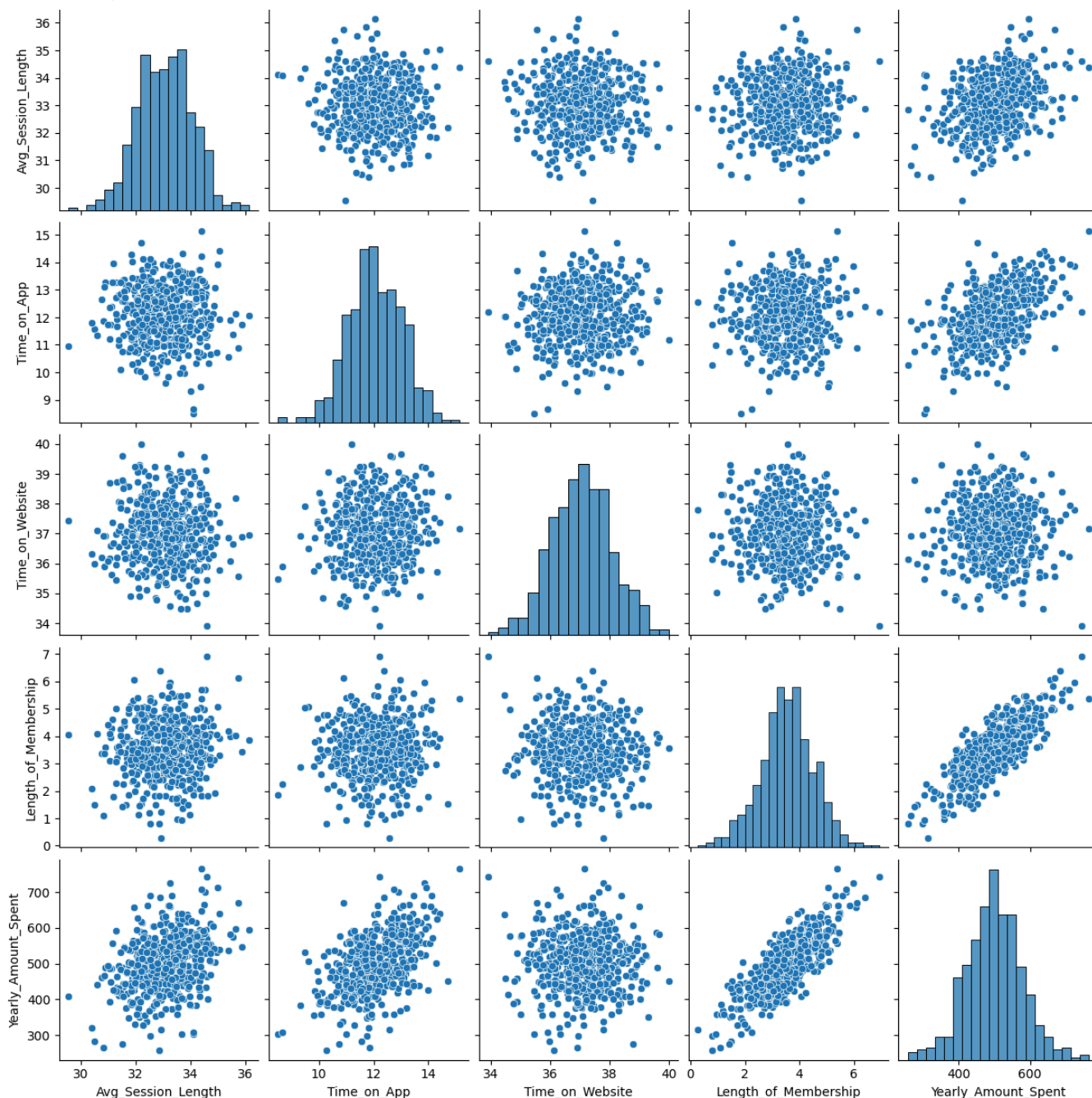
The correlation is less positive (greater than 0, but not nearer to 1). Spending time on app significantly increasing yearly spending.

Compare to the time on website, time on app is better to impact on yearly spending.

3. Compute -- Explore types of relationships across the entire data set using pairplot . Based off this plot what looks to be the most correlated feature with Yearly Amount Spent?

```
sns.pairplot(df)
```

 <seaborn.axisgrid.PairGrid at 0x7c4f1094a680>



```
correlation_matrix = df.drop(['Email', 'Address', 'Avatar'], axis=1).corr()
correlation_matrix['Yearly_Amount_Spent'].sort_values(ascending=False)
```

 **Yearly_Amount_Spent**

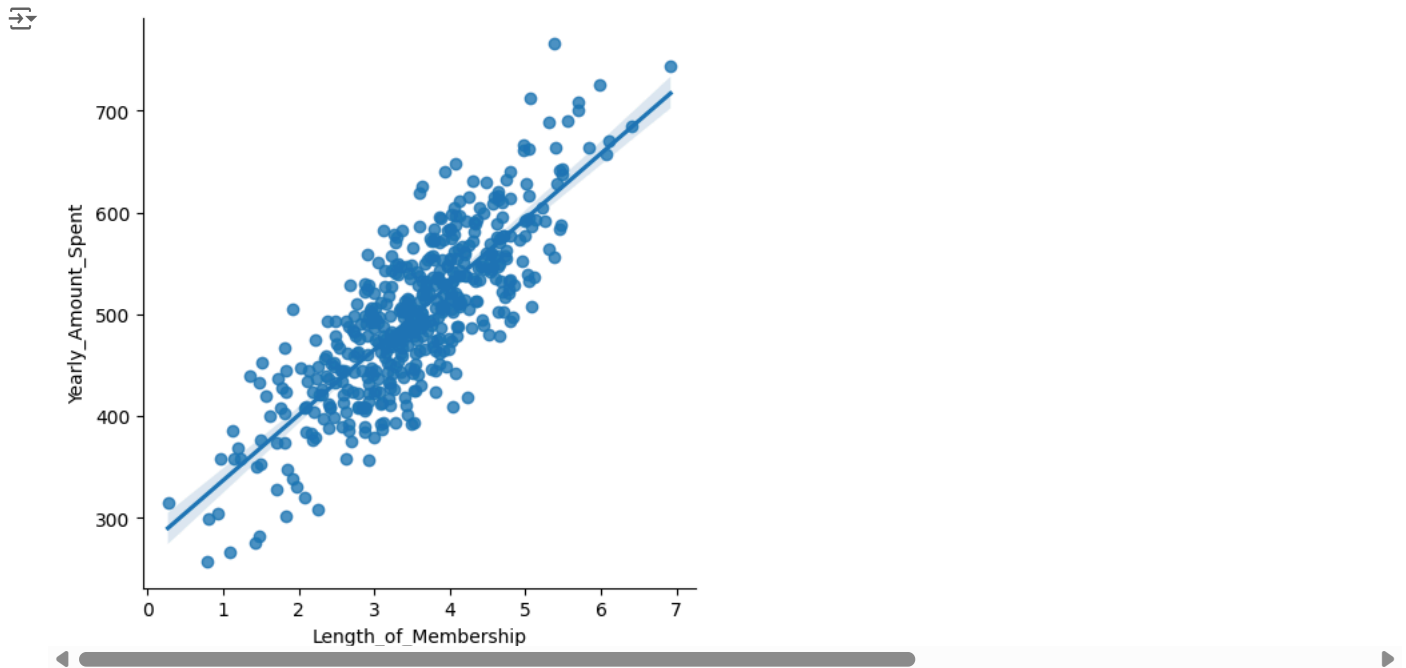
Yearly_Amount_Spent	1.000000
Length_of_Membership	0.809084
Time_on_App	0.499328
Avg_Session_Length	0.355088
Time_on_Website	-0.002641

data: 80x54

Length of Membership has the highest correlarion among others. It has positive correlation.

4. Compute – Create linear model plot of Length of Membership and Yearly Amount Spent. Does the data fits well in linear plot?

```
sns.lmplot(x='Length_of_Membership',y='Yearly_Amount_Spent',data=df)
plt.show()
```



The regression line fits well (points closely clustered around it), this suggests a strong relationship.

```
membership_corr = df['Length_of_Membership'].corr(df['Yearly_Amount_Spent'])
print(f"Correlation between Length of Membership and Yearly Amount Spent: {membership_corr}")
```

Correlation between Length of Membership and Yearly Amount Spent: 0.8090835682495691

*It suggests that longer memberships lead to higher yearly spending. *

5. Compute – Train and Test the data and answer multiple questions -- What is the use of random_state=85?

```
#Extracting features as x and vector as y.
x=df.drop(['Email','Address','Avatar','Yearly_Amount_Spent'],axis=1)
y=df['Yearly_Amount_Spent']

#Splitting to Train and Test Data.
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=85)
x_train.shape,x_test.shape,y_train.shape,y_test.shape
```

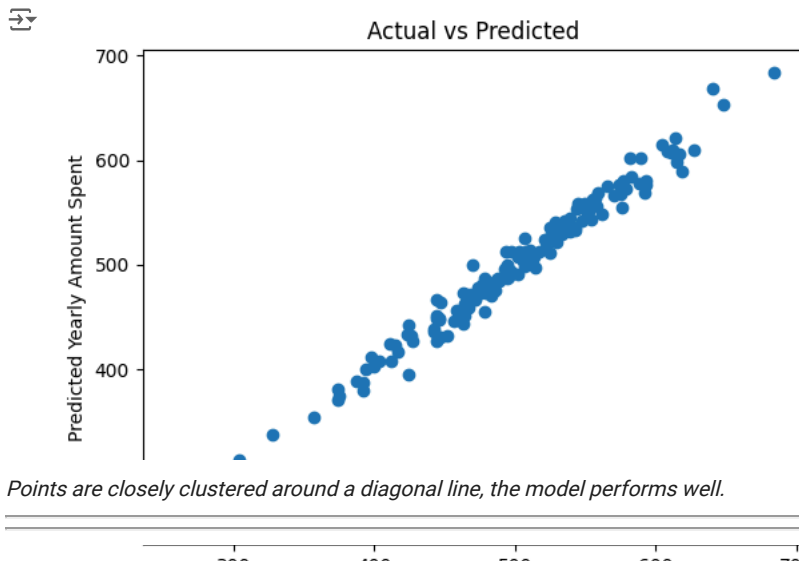
((350, 4), (150, 4), (350,), (150,))

6. Compute – Predict the data and do a scatter plot. Check if actual and predicted data match?

```
#Train the Model.
lr=LinearRegression()
lr.fit(x_train,y_train)

#Predicting the test data.
y_pred=lr.predict(x_test)

plt.scatter(y_test,y_pred)
plt.xlabel('Actual Yearly Amount Spent')
plt.ylabel('Predicted Yearly Amount Spent')
plt.title('Actual vs Predicted')
plt.show()
```



7. What is the value of Root Mean Squared Error?

```
mse=mean_squared_error(y_test,y_pred)
rmse=np.sqrt(mse)
rmse
```

10.572658672879333

8. Based on coefficients interpret company should focus more on their mobile app or on their website

```
coef=pd.DataFrame(lr.coef_,x.columns,columns=['Coefficient'])
```